Above the Ceiling

What Legal Meaning Preservation Metrics Actually Measure,
and How to Find Out

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Abstract

Automatic metrics for *legal* meaning preservation promise to tell a practitioner whether a simplified contract clause still says what the original said. This work is an extended version of JUDGEBERT and its companion corpus FRJUDGE [1], the first supervised metric and dataset for legal meaning preservation, carried further along four lines that its first release left open. Working from the raw annotations, we report four findings. First, the corpus is more reliable than first reported: the reported Krippendorff's α of 0.10 comes from applying a *nominal* coefficient to a ten-point *ordinal* scale; the appropriate ordinal coefficient is 0.325. Second, the metric's headline correlation sits *above* the human ceiling: a single annotator predicts the consensus of the other four at $r = 0.597$, and the best-agreeing pair of annotators reaches only $r = 0.606$, while the metric reports $r = 0.74$. Third, once every metric is calibrated to the human scale on the training split (a step the original comparison omits, which makes its RMSE column an artefact of supervision rather than of quality), a ridge regression over ten *surface* features (lengths, token overlap) reaches $r = 0.62$, matching the human ceiling and beating every neural baseline. Fourth, and most consequentially, we introduce LEGALEDIT, a diagnostic of 373 minimal perturbations that flip legal force (*doit* / *peut*, *et* / *ou*, *assuré* / *assureur*, dropped negations, amounts multiplied by ten) while preserving 0.93 of the tokens. It splits the field in two. Every similarity-based metric fails almost completely, spending 0.022–0.039 of its identical-to-unrelated range on an edit that reverses the law; on the items that swap *l'assuré* for *l'assureur*, and so reverse who owes the duty, they spend 0.020. Bidirectional NLI entailment, the strong baseline the original comparison omits, is the exception, spending 0.67 of its range and ranking 93% of items correctly. Crucially, the two rankings come apart: the best-correlating signal on the corpus is the worst on LEGALEDIT, so a metric chosen by correlation alone is chosen wrongly. We argue that the identical/unrelated sanity checks in current use are jointly solvable by lexical overlap and therefore cannot certify a legal metric, and propose LEGALEDIT as the missing third check. Retraining the metric under an evaluation that never scores it on its own augmentation also reverses one of the original results: the augmentation reported to raise r from 0.74 to 0.97 instead costs more than half of it, on every seed. A distributional target and a conservative quantile loss do not raise correlation either, but they cut the seed-to-seed spread by two thirds and deliver the risk asymmetry a task where five lawyers routinely differ by seven points on a ten-point scale actually calls for. Code, corrected splits, and the diagnostic are released.

1 Introduction

Simplifying a legal text is a rewriting problem with an unusual failure mode. When a news summary drops a qualifier the reader is mildly misinformed; when an insurance clause drops a qualifier the reader may be uninsured. The 2024 decision in *Moffatt v. Air Canada*, in which a tribunal found an airline liable for a chatbot's misstatement of its own bereavement-fare policy, illustrates that the gap between what a document says and what an automated system says it says is legally cognisable, not merely academic.

This motivates a distinct evaluation dimension. Beauchemin et al. [1] argue, convincingly in our view, that *legal* meaning preservation is not general meaning preservation: in ordinary French *automobile* and *véhicule* are near-synonyms, but under Quebec's Highway Safety Code an automobile is moved by mechanical force and a vehicle by mechanical *or human* force, so every bicycle is a counterexample. They contribute FRJUDGE, 297 clause/simplification pairs from Quebec property-insurance forms rated by five law students, and JUDGEBERT, a CamemBERTv2 regressor that reports a correlation with the mean human rating of $r = 0.74$, rising to 0.97 with data augmentation, against 0.46 for the strongest baseline. The construct and the corpus are the right foundation, and this work builds on them rather than around them; extending them at all is possible only because the raw annotations were released.

The question we take up is what the reported numbers license. A metric that will be used to certify legal text needs a specific kind of evidence: not that it correlates with an average of human ratings, but that it reacts to the things that change legal meaning and ignores the things that do not. We test that directly, and find a consistent pattern across four analyses.

The gold standard is weaker, and stronger, than reported. Reproducing the inter-annotator statistics exactly (our pairwise percent agreement matches the published 25.96 to two decimals), we find that the reported $\alpha = 0.10$ applies a nominal coefficient to an ordinal scale, treating a 9-versus-10 disagreement as identical to a 1-versus-10 disagreement. The appropriate ordinal coefficient is 0.325 (§3.1). At the same time the disagreement that remains is structural rather than random: the five annotators split into a lenient group (means 7.61) and a strict group (4.34), and on 63% of pairs their ratings span six points or more. The mean of five such ratings is a quantity no annotator asserted.

The reported correlation exceeds the human ceiling. Because the corpus ships five ratings per item, the ceiling is measurable: a held-out annotator predicts the mean of the other four at $r = 0.597$ (RMSE 3.27), and the best-agreeing *pair* of annotators reaches $r = 0.606$ (§4). A model reported at $r = 0.74$ is therefore not "better than the baselines at judging legal meaning"; it is better than any human at predicting a five-rater average, which is a different and much easier task, and one that rewards predicting the middle of a bimodal distribution. We show how to report this honestly, using the reliability of the aggregate to separate the two claims.

Surface features do most of the work. The original comparison scores unsupervised similarity metrics against a 1–10 Likert label without ever fitting them to that scale, so their RMSE measures the absence of supervision. After isotonic calibration on the training split, the ordering changes substantially, and a ridge regression over ten surface features (word-count difference, token overlap, numeral overlap) reaches $r = 0.62$, above every neural baseline and level with the human ceiling (§5). Legal meaning preservation, as operationalised by this corpus, is to a large extent length preservation.

A minimal legal edit separates the field in two. §6 introduces LEGALEDIT. The existing sanity checks pair a sentence with itself (expect 10) and with an unrelated sentence (expect 1). These vary lexical overlap and legal meaning *together*, so a pure overlap function passes both; this is why every embedding metric already passes the identical check. LEGALEDIT holds overlap fixed (0.93 token Jaccard) and moves legal meaning alone. Under that dissociation the similarity metrics collapse: BERTScore scores a sentence that says the opposite of the original at 0.982 where the original scores 1.000. Bidirectional NLI, the baseline the original study does not include, retains most of its discrimination. The diagnostic is therefore not an impossible bar: it is a bar that one existing family clears and the family JUDGEBERT belongs to does not.

Contributions. (i) A recomputation of the FRJUDGE annotation statistics from the raw export, and a correction to the reliability coefficient; (ii) the first human-ceiling estimate for legal meaning preservation, and a protocol for reporting metric quality relative to it; (iii) a calibrated baseline comparison including trivial surface features, showing most reported headroom is attributable to supervision and length; (iv) LEGALEDIT, a 373-item diagnostic that dissociates lexical overlap from legal force, on which every similarity-based metric fails and bidirectional entailment does not, and on which corpus correlation and legal sensitivity come apart; (v) a retraining study showing that the original augmentation, once held out of the evaluation set, costs more than half the reported correlation, and that disagreement-aware and risk-asymmetric objectives buy stability and calibration rather than accuracy; and (vi) a reproducibility and privacy review of the released artefacts.

2 Related Work

Evaluating text simplification. Simplification quality is conventionally decomposed into fluency, simplicity and meaning preservation. Metrics borrowed from translation and summarisation transfer poorly: Sulem et al. [2] show BLEU [3] does not correlate with meaning preservation once sentence splitting is involved, the operation most characteristic of legal simplification [4]. SARI [5] targets simplicity, not adequacy. Embedding metrics [6, 7] capture semantic variability better but, as Cripwell et al. [8] document, score consistently high across *all* alterations, penalising even substantial meaning changes too little. Trained metrics such as LENS [9] and MeaningBERT [10] improve correlation on the corpora they are fitted to. Agrawal and Carpuat [11] argue for grounding meaning preservation in reading comprehension rather than similarity, and Trienes et al. [12] characterise simplification errors as recoverable *information loss*, a framing close to the omission category that dominates legal error taxonomies.

Metric robustness and challenge sets. Our diagnostic follows a line of work showing that correlation on naturally occurring outputs hides brittleness on targeted perturbations. Sai et al. [13] build perturbation checklists for NLG metrics; Anschütz et al. [14] show metrics systematically miss negation; Chen and Eger [15] demonstrate that similarity-based metrics are not robust to meaning-changing edits and that NLI-based metrics repair much of the gap. LEGALEDIT adapts this method to a domain where the perturbations are not linguistic curiosities but the exact operations that decide coverage disputes. To our knowledge no prior challenge set targets legal force.

Factual consistency. Adequacy in summarisation is measured with entailment- and QA-based methods: SummaC [16], AlignScore [17], QAFactEval [18]. These are English-only, which is why Beauchemin et al. [1] could not use them; we therefore construct a French analogue by running a multilingual NLI model [19, 20] in both directions, so that failure of $o \Rightarrow s$ signals hallucination and failure of $s \Rightarrow o$ signals omission. This is, conceptually, the strongest baseline the original study is missing.

Annotator disagreement. The assumption that a single gold label exists has been extensively challenged [21, 22]. Uma et al. [23] survey methods that learn from the label distribution rather than its mode, and Peterson et al. [24] show soft labels improve robustness. Legal interpretation is a paradigm case: disagreement between competent lawyers is the ordinary condition of the field, not annotator error. We accordingly treat the five FRJUDGE ratings as a distribution to be modelled (§7) rather than averaged.

Meta-evaluation methodology. Deutsch et al. [25] show correlation estimates for generation metrics carry wide confidence intervals and that apparent metric differences frequently fail significance testing; Graham and Baldwin [26] give the appropriate test for dependent correlations; Card et al. [27] show NLP comparisons are routinely underpowered. We apply all three.

Dimension	Krippendorff's α		Reported
	nominal	appropriate	[1]
Simplicity level (nominal)	0.183	0.183	0.18
Characterization (nominal)	0.552	0.552	0.55
Legal meaning (**ordinal**)	0.104	**0.325**	0.10

Table 1: Krippendorff's α on FRJUDGE. Legal meaning is an ordinal ten-point scale; the published 0.10 is the nominal coefficient, which is not the appropriate measure for this design.

3 Corpus and Reconstruction

FRJUDGE contains 297 pairs drawn from two Quebec insurance forms (one BAC home-insurance form, the AMF automobile forms), each simplified by `gpt-4-turbo-2024-04-09` with a zero-shot prompt and rated by five law students on legal meaning (1–10), simplicity (4 levels), and an 18-class legal characterization.

Two files are released. `FrJUDGE.jsonl` contains 1,485 rows (297×5) but *drops annotator identity*, making annotator-level analysis impossible. The raw Prodigy export retains it in `_annotator_id`. We rebuild the corpus from the raw export, recovering the five annotators (pseudonymised A–E here; see §12), the source document of each clause, and per-annotation timestamps. We verify the reconstruction against the published statistics: our mean pairwise percent agreement is 25.96%, matching the published value exactly, and our recovered per-annotator means (4.33, 7.57, 7.55, 7.72, 4.36) match the published Figure 2c to within rounding.

3.1 The reliability coefficient

Table 1 resolves a discrepancy. Our nominal α for legal meaning is 0.104, reproducing the published 0.10; our nominal α for simplicity (0.183) and characterization (0.552) likewise reproduce the published 0.18 and 0.55. But legal meaning is measured on an ordered ten-point scale, and nominal α counts a one-point disagreement as fully as a nine-point one. With the ordinal coefficient appropriate to the design [28, 29] the value is 0.325.

This matters in both directions. The corpus is more reliable than the published coefficient makes it look, which is a point in its favour that the original figure understates and a reader would otherwise miss. But 0.325 still indicates only fair agreement, and the disagreement is not noise.

3.2 The structure of disagreement

The five annotators separate into two stable populations (Figure 1a): B, C and D average 7.61 and A and E average 4.34, a gap of over three points on a ten-point scale. Ratings are strongly bimodal (the two modal values are 10 and 1), the median across-annotator range on a single pair is 7 points, and 63% of pairs span at least six points (Figure 1b). Averaging produces a target with a standard deviation of only 2.40, substantially narrower than the underlying ratings, which is precisely the condition under which a regressor that predicts the middle scores well.

4 The Human Ceiling

Because every item carries five ratings, the ceiling is directly measurable. For each annotator we correlate their ratings against the mean of the other four (Table 2). The mean leave-one-annotator-out correlation

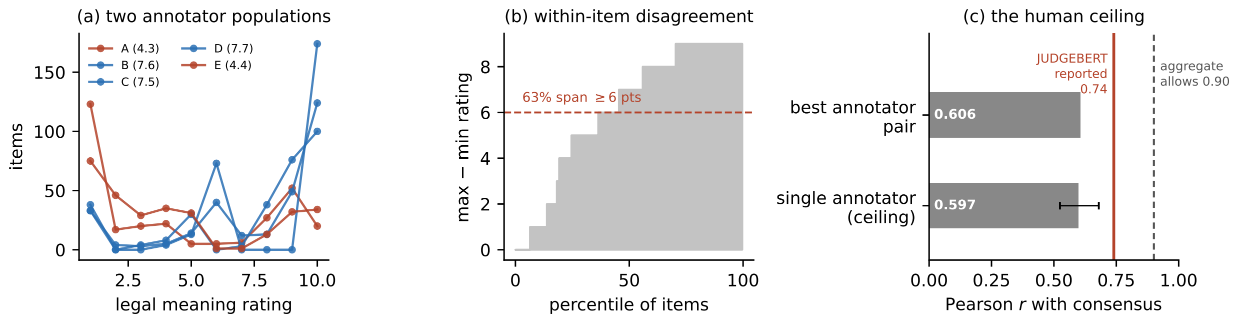

Figure 1: Disagreement in FRJUDGE. (a) The five annotators form two populations: B, C, D (blue) rate legal preservation about three points higher than A and E (red) across the whole corpus. (b) Sorted within-item range; the median item spans 7 points of a ten-point scale. (c) A single annotator predicts the consensus of the other four at $r = 0.597$ (bar; whiskers give the range over the five held-out annotators), and the best-agreeing pair reaches $r = 0.606$; both sit below the 0.74 reported for JUDGEBERT.

Held-out annotator	r (95% CI)	ρ	RMSE
A	0.524 [0.44, 0.60]	0.551	3.93
B	0.569 [0.48, 0.65]	0.586	3.05
C	0.680 [0.60, 0.75]	0.725	2.75
D	0.601 [0.52, 0.67]	0.557	3.08
E	0.612 [0.55, 0.67]	0.679	3.56
Mean (human ceiling)	**0.597**	0.620	3.27
Best annotator pair (C-D)	0.606	–	–
Mean pairwise	0.462	–	–

Table 2: Leave-one-annotator-out ceiling on FRJUDGE ($n = 297$). A single expert predicts the consensus of the other four at $r = 0.597$. JUDGEBERT is reported at $r = 0.74$.

is 0.597 (RMSE 3.27); the best-agreeing pair of annotators reaches $r = 0.606$ and the mean pairwise correlation is 0.462.

How to read a metric that beats it. A model exceeding 0.597 is not thereby superhuman. Predicting the *mean* of five raters is easier than being one of them, because averaging cancels idiosyncratic variance that no individual can cancel for themselves. The appropriate upper bound for predicting the observed mean is set by the reliability of that mean, which by Spearman–Brown from the mean pairwise correlation is 0.811; a metric can in principle approach $\sqrt{0.811} = 0.90$. Reporting both bounds separates two claims that the original paper conflates: *as good as an expert* ($r \approx 0.597$) and *as good as the aggregate allows* ($r \approx 0.90$). We recommend that any future legal-meaning metric report its correlation against both.

5 Calibrated Baselines

5.1 Why the original comparison is not decidable

The original study normalises each metric by decimal scaling into $[0, 10]$ and then reports RMSE against a 1–10 human label. Decimal scaling is not calibration: it fixes the range but not the location or shape of the distribution. An unsupervised cosine similarity that happens to occupy $[0.7, 1.0]$ will map to high scores everywhere and record a large RMSE no matter how well it ranks. JUDGEBERT, by contrast, is

Metric	r	ρ	$\mathrm{RMSE}_{\mathrm{raw}}$	$\mathrm{RMSE}_{\mathrm{cal}}$	over%
Unsupervised, French-capable					
BERTScore-CamemBERTv2	$0.291_{\pm 0.09}$	0.326	2.62	2.36	49.3
BERTScore-CamemBERTv2-Recall	$0.352_{\pm 0.12}$	0.398	2.59	2.30	47.2
BERTScore-FlauBERT	$0.482_{\pm 0.08}$	0.459	2.16	2.12	49.4
LaBSE	$0.409_{\pm 0.06}$	0.402	2.33	2.21	47.2
Sentence-CamemBERT	$0.296_{\pm 0.05}$	0.287	2.68	2.31	49.3
Para-mUSE-mpnet	$0.252_{\pm 0.08}$	0.266	2.88	2.35	47.2
mE5-base	$0.358_{\pm 0.08}$	0.356	2.52	2.26	48.0
NLI-fwd (no hallucination)	$0.368_{\pm 0.06}$	0.352	3.85	2.25	49.3
NLI-bwd (no omission)	$0.340_{\pm 0.09}$	0.355	3.70	2.28	46.9
NLI-bidirectional (min)	$0.395_{\pm 0.07}$	0.405	3.74	2.22	46.6
Trivial surface features					
$-\lvert\,\lvert s\rvert - \lvert o\rvert\,\rvert$ (length difference)	$0.641_{\pm 0.04}$	0.624	3.25	1.85	47.0
Token Jaccard	$0.303_{\pm 0.06}$	0.331	3.34	2.34	48.0
Length ratio	$0.580_{\pm 0.05}$	0.560	3.43	1.97	49.1
Surface-Ridge (10 features, supervised)	$0.621_{\pm 0.05}$	0.588	1.89	1.89	45.8
Human ceiling (one annotator)	**0.597**	0.620	–	3.27	–

Table 3: Calibrated comparison on the FRJUDGE test split, mean over 10 seeds. $\mathrm{RMSE}_{\mathrm{raw}}$ is the original decimal-scaling protocol; $\mathrm{RMSE}_{\mathrm{cal}}$ fits an isotonic map on the training split. r and ρ are unaffected by calibration for the unsupervised metrics.

trained on the label itself. The reported RMSE column therefore measures access to supervision, and the companion statistic (the fraction of predictions above the human label, on which JUDGEBERT scores 0% and BERTScore 82%) follows mechanically from the same asymmetry.

We repair this by fitting an isotonic regression from each metric's raw score to the human scale on the training split only, and evaluating on the held-out test split, so that every metric is compared under identical supervision. Isotonic calibration is monotone, so it cannot improve a metric's ranking of items; it only removes the scale artefact [30].

5.2 Baselines

We replace the original English-only comparison set with French-capable metrics: BERTScore [6] over CamemBERTv2 [31] and FlauBERT [32], both F1 and recall-only, since recall directly targets omission; sentence-embedding cosine with LaBSE [33], Sentence-CamemBERT, multilingual mpnet, and multilingual E5; and bidirectional NLI entailment with mDeBERTa-XNLI [19]. We add trivial surface features and a supervised ridge regression over ten of them.

Table 3 reports the result. Three observations. First, calibration collapses most of the reported RMSE gap: the metrics were not as far from human judgment as the original table suggests, they were merely on a different scale. Second, the strongest single unsupervised signal is not a semantic one. Third, a ridge regression over word counts and token overlap reaches $r = 0.62$, which is at the human ceiling and above every neural baseline. Whatever JUDGEBERT adds beyond that, the original paper does not measure, because it reports no such baseline.

We do not read this as evidence that the corpus is trivial. We read it as evidence about what the *annotation* rewards: under the published rubric, score is reduced once per identified error, and the most frequent error type in LLM simplification is omission, which is strongly correlated with becoming shorter. A metric can therefore score well by detecting deletion, without any representation of legal force. §6 tests exactly

Neural metric m	$r(m,h)$	$r(m,\ell)$	t	p
BERTScore-CamemBERTv2	0.344	0.349	5.05	<0.001
BERTScore-CamemBERTv2-Recall	0.396	0.443	4.43	<0.001
BERTScore-FlauBERT	0.484	0.389	2.61	0.010
LaBSE	0.424	0.362	3.66	<0.001
Sentence-CamemBERT	0.222	0.148	6.33	<0.001
Para-mUSE-mpnet	0.225	0.159	6.32	<0.001
mE5-base	0.318	0.316	5.40	<0.001
NLI-fwd (no hallucination)	0.320	0.388	5.66	<0.001
NLI-bwd (no omission)	0.307	0.258	5.36	<0.001
NLI-bidirectional (min)	0.365	0.350	4.68	<0.001

Table 4: Williams tests against the trivial length baseline $\ell = -\left|\,|s| - |o|\,\right|$, which reaches $r = 0.613$ with the human label h over all 297 items. $r(m,\ell)$ is each metric's correlation with the length feature itself, the dependence the test corrects for. Positive t favours ℓ.

that.

5.3 Statistical adequacy

The test split holds 89 items. At that size, the smallest correlation reliably distinguishable from $r = 0.46$ at 80% power is 0.73; the distance between the published 0.74 and 0.46 is therefore at the edge of detectability, and no finer comparison in the original table is resolvable. We report bootstrap confidence intervals throughout, following Deutsch et al. [25], and use the Williams test for dependent correlations [26] when comparing metrics against the same human scores.

Table 4 applies that test to the comparison the previous subsection turns on. Every neural metric and the length feature correlate with the *same* human label on the *same* 297 items, and with each other, so the comparison is dependent and a two-sample test on Fisher-transformed correlations would be anticonservative. Under Williams, the length feature's advantage is significant against every metric in the comparison set. We stress what this does and does not show: it establishes that the trivial baseline is genuinely stronger *on this corpus*, not that the neural metrics reason less well about law. §6 is what separates those two readings.

6 LegalEdit: Holding Overlap Fixed

6.1 Why the existing sanity checks cannot certify a legal metric

Two automated checks are currently used [1, 10]: an identical pair must score 100%, an unrelated pair 0%. Both are necessary. Neither, nor their conjunction, is close to sufficient, and the reason is structural: moving from an identical pair to an unrelated pair changes lexical overlap and legal meaning *in the same direction at the same time*. Any monotone function of token overlap passes both. This is precisely what the original results show: BERTScore, SBERT and MeaningBERT all pass the identical check, and pass it for the trivial reason that they are overlap functions of one kind or another.

6.2 Construction

LegalEdit breaks the confound. We take statutory sentences held out from FrJudge (the Quebec Automobile Insurance Act and the Highway Safety Code) and apply a single minimal edit that changes legal force while leaving the sentence otherwise intact:

peut → doit	*tier 1*
o: La Société peut refuser une indemnité, en réduire le montant, en suspendre ou en cesser le paiement dans les cas suivants: 1.	
s': La Société **doit** refuser une indemnité, en réduire le montant, en suspendre ou en cesser le paiement dans les cas suivants: 1.	
doit → peut	*tier 1*
o: Le candidat doit toutefois réussir les examens de compétence visés à l'article 67 pour obtenir un permis de conduire pour motocyclette.	
s': Le candidat **peut** toutefois réussir les examens de compétence visés à l'article 67 pour obtenir un permis de conduire pour motocyclette.	
drop *ne … pas*	*tier 1*
o: Les feux visés au paragraphe 2ř ne sont pas requis pour tout véhicule dont la largeur excède 2,03 mètres.	
s': Les feux visés au paragraphe 2ř sont requis pour tout véhicule dont la largeur excède 2,03 mètres.	
avant ↔ après	*tier 1*
o: Elle n'est pas due avant le septième jour qui suit celui de l'accident, sauf dans le cas prévu au troisième alinéa de l'article 57.	
s': Elle n'est pas due **après** le septième jour qui suit celui de l'accident, sauf dans le cas prévu au troisième alinéa de l'article 57.	
tous les → certains	*tier 1*
o: Un agent de la paix peut enlever ou faire enlever aux frais du propriétaire toute chose utilisée en contravention au présent article.	
s': Un agent de la paix peut enlever ou faire enlever aux frais du propriétaire **certaine** chose utilisée en contravention au présent article.	

Table 5: LEGALEDIT samples, one per perturbation rule, with the edited span in bold. Each remains fluent and statutory in register, shares almost all of its tokens with the original, and states a different rule of law.

- **Modality**: *doit ↔ peut*; obligation becomes permission.
- **Party**: *l'assuré ↔ l'assureur*; the duty changes hands.
- **Polarity**: dropping *ne … pas*; *couvre ↔ ne couvre pas*; *exclut ↔ inclut*.
- **Quantum**: the first monetary or temporal amount multiplied by ten (*12 mois → 120 mois*).
- **Scope**: *tous les → certains*; *sauf si → si*.
- **Temporal**: *avant ↔ après*.
- **Class** (tier 2): *automobile ↔ véhicule*, the original paper's own example; and *et ↔ ou*, which converts conjunctive conditions to disjunctive ones.

We separate a *tier 1* set, where the edit unambiguously changes who is bound, what is covered, or how much, from a *tier 2* set of scope shifts, and report both. Perturbations that introduce a grammatical artefact are filtered by a conservative French well-formedness check covering failed elision (*le automobile*), missing contraction (*de le*), doubled negation and determiner–noun gender disagreement, so that a metric cannot detect the edit by noticing disfluency instead of by reading the law. The check flags none of the 400 unmodified source sentences, so it removes only genuine artefacts. The result is 373 pairs (310 tier 1) with mean token Jaccard 0.93 against their source, none below 0.75. Table 5 gives one example per rule.

6.3 The diagnostic

For each source sentence s with perturbation s' we compare a metric's score on (s, s), on (s, s'), and on (s, u) for an unrelated legal sentence u. A metric that measures legal meaning should place (s, s') near

(s, u); a metric that measures overlap will place it near (s, s). We summarise with the *margin fraction*

$$\text{margin} \;=\; \frac{\overline{m(s,s)} - \overline{m(s,s')}}{\overline{m(s,s)} - \overline{m(s,u)}},$$

the share of the metric's identical-to-unrelated dynamic range that it spends on a legally decisive edit. A perfect legal metric scores 1; a pure overlap function scores near 0.

6.4 Results: two families, not one

Table 6 and Figure 3 report the diagnostic over 12 scorers, and the result is sharper than we expected. The metrics divide cleanly into two families.

Every *similarity*-based metric fails, and fails almost completely. The seven embedding and BERTScore variants spend between 0.022 and 0.039 of their working range on an edit that reverses the law: BERTScore over CamemBERTv2 places a sentence saying the opposite of the original at 0.982 on a scale where the original itself sits at 1.000 and an unrelated statute sits at 0.193. Token Jaccard, at 0.079, is in the same regime, which is the point: these metrics are behaving as overlap functions, exactly as the argument of §6 says they must. Their *discrimination* rates are uninformative for the same reason: they rank the identical pair above the edited one 100% of the time, but by a margin so small that it carries no decision.

Bidirectional NLI does not fail. It spends 0.67 of its range on the edit, ranks correctly on 93.3% of items, and places a legally flipped sentence (0.265) nearer the unrelated pair (0.010) than the identical one (0.784). The two directions individually reach 0.424 and 0.473; taking their minimum, which is what an omission-*or*-hallucination reading of the construct requires, is what produces the 0.670. This is the baseline the original comparison omits, and on the diagnostic that matters for the construct it is not merely competitive but in a different class.

Relation to MENLI. That entailment survives meaning-changing edits better than similarity is not a new claim: Chen and Eger [15] establish it for general-domain NLG metrics, and our result is in that sense a confirmation rather than a discovery. We think the confirmation is worth having, since it was not obvious that it would transfer to a domain where the decisive edits are single function words in statutory French, nor that the gap would be this large. But we want to be clear about what is new here and what is not. Three things are new. The perturbation inventory is defined by legal effect rather than by linguistic category, so the diagnostic speaks to a deployment question (may this metric gate a simplified clause?) rather than to metric robustness in general. The dissociation reported above, that corpus correlation does not order metrics the same way legal sensitivity does, is a property of FRJUDGE and could not be read off MENLI. And the observation in the next paragraph, that the sanity check this literature has standardised on would have excluded the entailment family outright, is specific to how legal-meaning metrics are currently validated.

The existing checks would have rejected it. Note where NLI's identical score sits: 0.784, not 1.000. This is not a normalisation artefact. In raw terms mDeBERTa assigns a mean entailment of 0.780 to a sentence paired with *itself*, and clears the 99% bar that Beauchemin et al. [1] require on only 1.3% of items, against 100% for every BERTScore and embedding metric. The metric that best tracks legal meaning is therefore the one metric that would be *disqualified* by the sanity check currently in use. We think this is the strongest available argument for the position of §6: the identical-pair check as presently applied does not merely fail to identify a legal metric, it actively selects against the family that has the property being sought. A metric that saturates on identical input is easy to build, since every overlap function does it by construction, and saturation is evidence about a metric's calibration at one endpoint, not about whether it reads the law. It should be reported, and it should not be a filter.

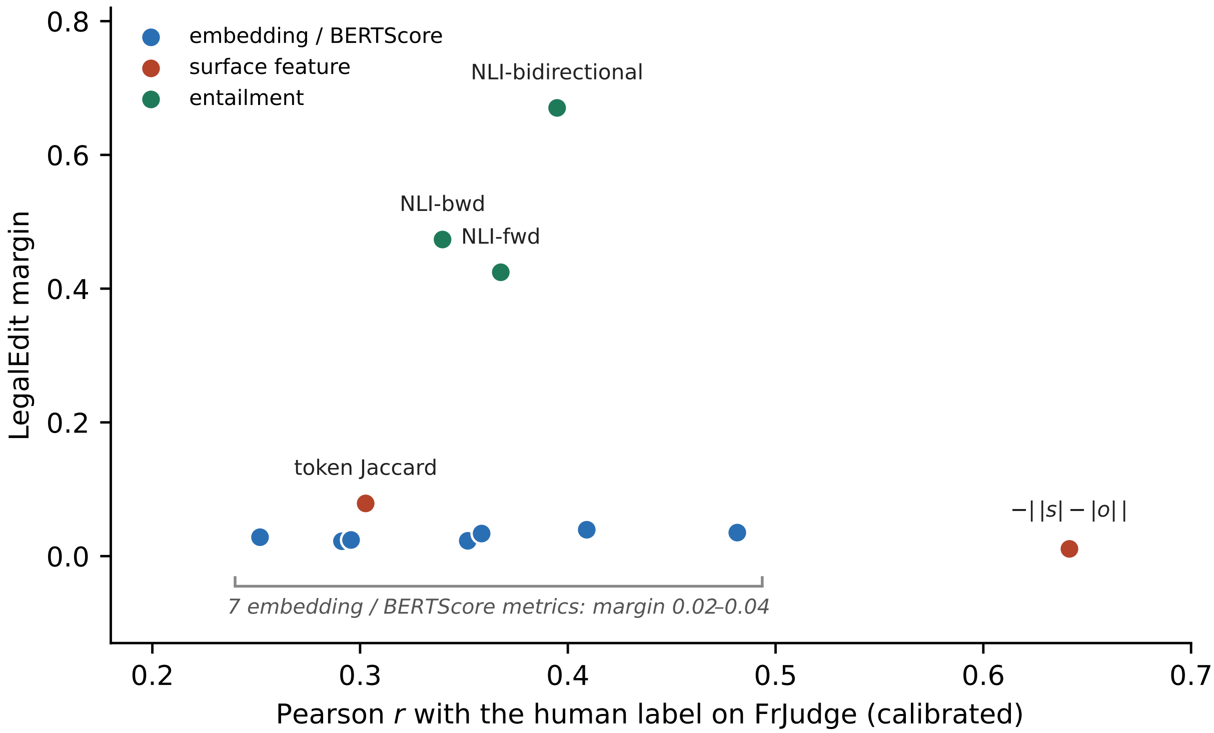

Figure 2: The two orderings come apart. Horizontally, how well a metric correlates with the human label on FRJUDGE after calibration, the quantity the original comparison selects on. Vertically, the share of its working range it spends on an edit that reverses the law. Selecting on the horizontal axis picks the length feature, at the bottom right, which is the worst scorer on the vertical one; the only family that moves at all sits in the middle of the horizontal axis.

The two rankings do not agree with each other. Over the 10 neural baselines the rank correlation between calibrated r on FRJUDGE and LEGALEDIT margin is $\rho = 0.58$, with a 95% bootstrap interval of $[-0.08, 0.90]$. At this number of metrics, correlation with the human label simply does not predict legal sensitivity, in either direction. The extremes show why it matters: the trivial length feature is the best-correlating signal on the corpus ($r = 0.641$, Table 3) and has the lowest margin of anything we test (0.011, and it ranks the edited pair correctly only 11.3% of the time), while bidirectional NLI sits mid-table on correlation ($r = 0.395$) and is the only family that reacts to the edit at all. A metric selected on FRJUDGE correlation alone would pick the wrong one. Figure 2 plots the two axes against each other.

Is the failure uniform across edit types? Table 7 breaks the margin down by perturbation rule, averaged over metrics. No rule is reliably detected: the best-detected operation, dropping *ne . . . pas*, still averages only 0.333, and it is plausibly the easiest because it is the one rule that *removes* tokens rather than substituting them. At the other end sits the edit with the most direct legal consequence in an insurance contract. Swapping the bound party, *l'assuré* for *l'assureur*, which reverses who owes the duty and who is owed it, draws a mean margin of 0.046 in one direction and -0.005 in the other. Those two cells rest on 11 and 3 items respectively, so we pool them: over all 14 party-swap items the seven similarity metrics spend 0.020 of their range (per-metric range 0.006–0.036), against a mean of 0.155 across all rules. Reversing which party a clause binds is close to the least detectable thing one can do to it. We tested whether the residual ordering is explained by the size of the edit in characters and found no relationship ($r = 0.25$ over the 12 rules, 95% CI $[-0.66, 0.73]$). With 12 rules of unequal frequency the ordering is not itself well determined,

Metric	identical	LEGALEDIT	unrelated	discrim.%	margin
NLI-bidirectional (min)	0.784	0.265	0.010	93.3	0.670
NLI-bwd (no omission)	0.784	0.432	0.042	77.2	0.473
NLI-fwd (no hallucination)	0.784	0.469	0.043	77.7	0.424
surface: token Jaccard	1.000	0.928	0.089	99.2	0.079
LaBSE	1.000	0.972	0.292	100.0	0.039
BERTScore-FlauBERT	1.000	0.979	0.393	100.0	0.035
mE5-base	1.000	0.975	0.266	100.0	0.034
Para-mUSE-mpnet	1.000	0.981	0.312	100.0	0.028
Sentence-CamemBERT	1.000	0.983	0.316	100.0	0.024
BERTScore-CamemBERTv2-Recall	1.000	0.983	0.240	100.0	0.023
BERTScore-CamemBERTv2	1.000	0.982	0.193	100.0	0.022
surface: -|len diff|	1.000	0.997	0.718	11.3	0.011

Table 6: LEGALEDIT. Scores min–max normalised per metric. *discrim.* is the share of items ranked correctly (identical $>$ edited); *margin* is the fraction of the identical-to-unrelated range spent on the legal edit. Correct ranking is necessary but far from sufficient: the similarity metrics rank 100% of items correctly while moving almost nothing, which is why the margin, not the rate, is the quantity to read.

and we report it descriptively; the robust observation is the ceiling, not the ranking.

7 Disagreement-Aware and Risk-Aware Metrics

The gold label is the mean of five ratings whose median spread is 7 points. Fitting a scalar regressor to that mean asks the model to predict a quantity none of the annotators asserted, and rewards predicting the centre. We therefore train five variants on identical splits, differing only in the training target and loss:

- **Scalar** (the original JUDGEBERT objective): MSE against the mean rating.
- **Distributional**: a ten-way head trained with KL divergence against the empirical rating histogram, following Uma et al. [23] and Peterson et al. [24]. The reported score is the distribution's expectation, so it is directly comparable, but the model additionally exposes its spread.
- **Annotator-aware**: five regression heads, one per annotator, averaged at inference. This lets the model represent the lenient and strict populations separately instead of averaging them away.
- **Quantile**: a pinball loss [34] at $\tau = 0.25$. The asymmetry is deliberate. As Beauchemin et al. [1] correctly observe, a metric that *overshoots* the human rating is permissive of bad simplifications, while one that undershoots is merely strict, and in a legal setting those errors are not symmetric. A quantile objective encodes that preference in the loss rather than hoping it emerges.
- **Multi-task**: the scalar head plus an auxiliary characterization head, trained against the majority vote of the five annotators, which spans 17 of the rubric's 18 classes (the eighteenth never carries a majority). The characterization labels are collected in FRJUDGE and then discarded; since the published rubric has annotators characterize a clause *before* scoring it, the label plausibly carries information about the score.

We additionally train two augmented variants: **DA**, the original identical/unrelated augmentation carried over unchanged, and **DA+LegalEdit**, which adds minimally-edited negatives to the training mixture.

Perturbation	n	mean margin	worst metric
drop *ne ... pas*	21	0.333	0.032
avant $\leftrightarrow$ *après*	9	0.272	0.006
amount $\times$ 10	21	0.256	0.025
doit $\rightarrow$ *peut*	82	0.241	0.015
sauf si $\rightarrow$ *si*	10	0.225	0.014
peut $\rightarrow$ *doit*	111	0.167	0.018
tous les $\rightarrow$ *certains*	42	0.153	0.017
et $\rightarrow$ *ou*	27	0.069	0.007
ou $\rightarrow$ *et*	26	0.069	0.004
assureur $\rightarrow$ *assuré*	11	0.046	0.007
automobile $\leftrightarrow$ *véhicule*	10	0.027	-0.025
assuré $\rightarrow$ *assureur*	3	-0.005	-0.086
All	373	**0.155**	–

Table 7: LEGALEDIT margin by perturbation rule, averaged over the metrics in Table 6. *worst metric* is the lowest margin any single metric assigns that rule. A margin of 1 would mean the metric treats a legally flipped sentence exactly as it treats an unrelated one.

Evaluation protocol. Unlike the original study, correlation and RMSE are computed *only* on the human-annotated test split, never on a split containing augmentation. This matters: adding identical pairs (label 10) and unrelated pairs (label 1) to the evaluation set inflates label variance at the extremes, which raises Pearson r for every metric without any improvement in judgment. In the original Table 5 this inflation moves BERTScore from 0.46 to 0.94 while its RMSE simultaneously *worsens* from 3.61 to 5.09, a signature of variance inflation rather than of better measurement. Augmentation here affects training only; the three diagnostic regimes are reported separately as pass rates.

The training target barely moves the correlation. Table 8 reports the grid. The first result is a negative one. The distributional head gains 0.030 of Pearson r over the scalar baseline and the quantile head 0.011, while the annotator-aware and multi-task heads lose 0.008 and 0.023. Every one of those differences is smaller than the scalar baseline's own spread across seeds (± 0.082), and on paired seeds the best of them wins three times out of five. §5 put the smallest resolvable gap on a 89-item test split at roughly a quarter of a correlation point; that applies to our variants as much as to the original's baselines, and we report the ordering without asking the reader to believe it.

The differences that do survive are in the other columns. The distributional head is markedly steadier than the scalar head it replaces, ± 0.029 against ± 0.082 over the same five seeds, reaches the best RMSE in the table (1.99), and roughly doubles the rank correlation ($\rho = 0.578$ against 0.280), which is what one expects from a target carrying the shape of the rating distribution and not only its mean. The quantile head does precisely what it was asked to do: it overshoots the human label on 9.9% of test items against 25.4% for the scalar head, the lowest rate of any model here. The risk asymmetry Beauchemin et al. [1] argue for can therefore be written into the loss rather than hoped for, and it costs nothing in correlation. We read the disagreement-aware objectives as buying stability and calibration, not accuracy.

Augmentation, scored honestly, is harmful. The two augmented mixtures are the clearest effect in the table and they run against the original finding. JUDGEBERT-DA loses 0.350 of correlation against the un-augmented baseline and DA+LEGALEDIT loses 0.420, on five seeds out of five in both cases, and the instability is severe: DA varies by ± 0.254 across seeds and reaches $r = -0.188$ on one of them. The original paper reports augmentation raising r from 0.74 to 0.97. The difference is entirely the evaluation set. That

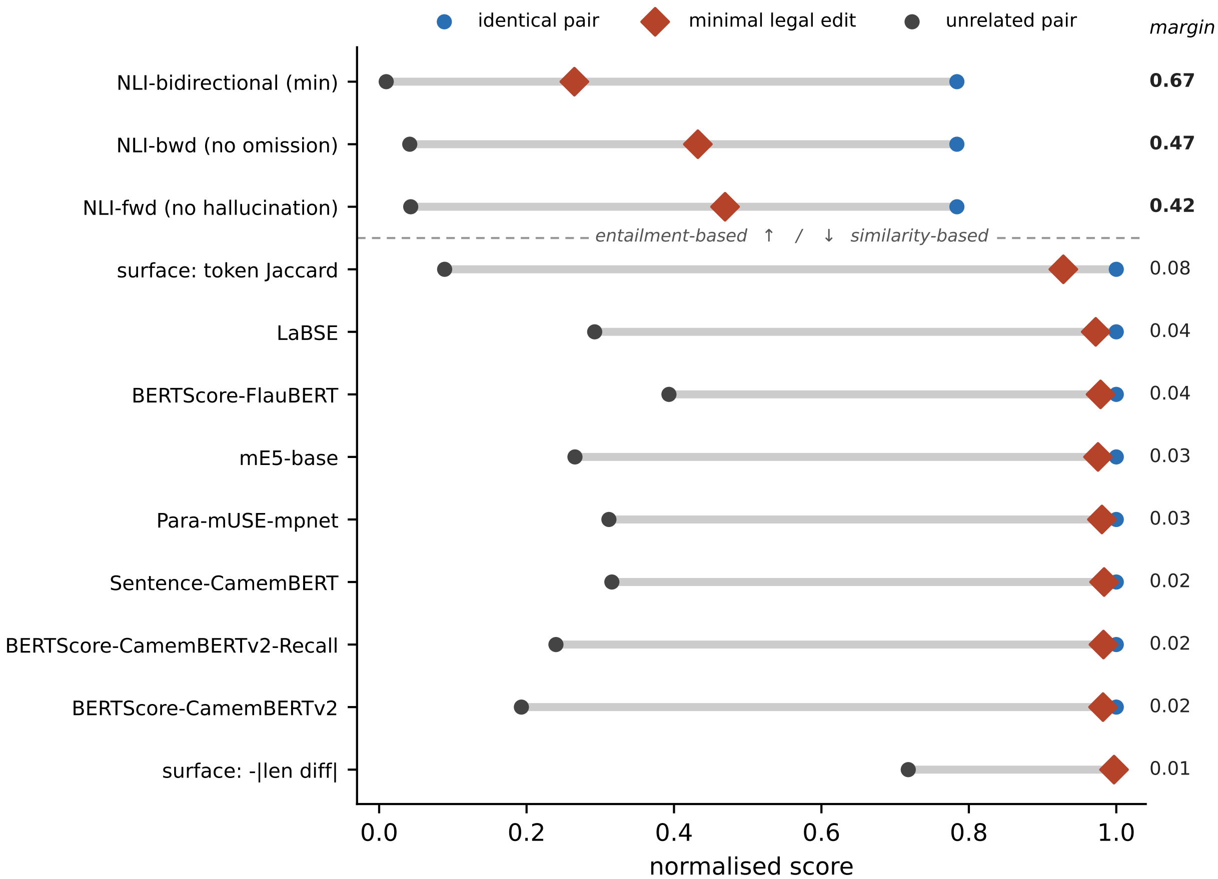

Figure 3: LEGALEDIT. For each metric the grey bar spans its score on an unrelated pair (dark circle) to its score on an identical pair (blue circle): its full working range. The red diamond is where it places a sentence whose legal force has been reversed by a single word. A metric that measured legal meaning would put the diamond at the left end. Every similarity-based metric puts it at the right, hard against the identical pair; the three NLI variants, at the top, are the only ones that move it appreciably.

figure is computed on a split containing the augmentation, where identical pairs at 10 and unrelated pairs at 1 supply most of the label variance; scored on the human-annotated items alone, the same augmentation costs more than half the correlation. This is the concrete form of the variance-inflation argument above, and it is the reason we separate the two evaluations rather than reporting a single number.

Trained constraints do not cross a change of document. Every model in Table 8 scores 0% on the identical-pair check, the two augmented ones included, although identical pairs are in their training mixture. The unrelated check goes the other way: the augmented models pass it on 77% and 69% of probes and the un-augmented models on none. The asymmetry is exactly the one our probe design predicts. Our unrelated probes are drawn from the same two statutes the training mixture draws its unrelated pairs from, so passing that check is an in-distribution result; our identical probes are statutory sentences while the mixture's identical pairs are FRJUDGE clauses, so passing that one would require generalising across source documents, and no model does. The same holds for the diagnostic: DA+LEGALEDIT, trained on minimally-edited negatives, reaches a margin of 0.051, against 0.670 for untrained bidirectional NLI. Its negatives are generated from FRJUDGE originals and its probes from statutes, and the sensitivity does not survive the move. One column here should be read with care: LEGALEDIT false-acceptance is the share of flipped

Model	r	ρ	RMSE	over%	ident.	unrel.	LEGALEDIT	
					pass	pass	false-acc.	margin
JudgeBERT-Scalar (baseline)	$0.542_{\pm 0.08}$	0.280	2.48	25.4	0	0	0	–
JudgeBERT-Dist (soft labels)	$0.572_{\pm 0.03}$	0.578	1.99	42.9	0	0	69	0.015
JudgeBERT-Annot (5 heads)	$0.534_{\pm 0.05}$	0.585	3.47	12.6	0	0	0	–
JudgeBERT-Quantile (tau=.25)	$0.553_{\pm 0.06}$	0.377	3.28	9.9	0	0	0	–
JudgeBERT-MT (+charact.)	$0.519_{\pm 0.06}$	0.232	2.51	26.3	0	0	0	–
JudgeBERT-DA	$0.192_{\pm 0.25}$	0.112	2.31	42.5	0	77	18	0.019
JudgeBERT-DA+LegalEdit	$0.122_{\pm 0.10}$	0.174	2.60	35.3	0	69	0	0.051
NLI-bidirectional (untrained)	0.395	0.405	2.22	46.6	1	–	–	*0.670*
Human ceiling	**0.597**	0.620	3.27	–	–	–	–	–

Table 8: Trained metrics, mean over 5 seeds, evaluated on the human-annotated test split only. *ident.* and *unrel.* are the original two sanity checks; LEGALEDIT false-acceptance is the share of legally flipped pairs still scored ≥ 7 ("accurate" under the published rubric). The margin is the quantity defined in §6 and reported for the untrained metrics in Table 6. It is shown as '–' where a model places the identical and unrelated probes within one point of each other, because a model with no working range has none to spend on the edit, and the ratio of two such differences carries no information.

pairs still scored ≥ 7, so it is confounded with where a model puts its predictions overall. The distributional head's 69% reflects a head that scores high in general (its identical probes average 7.22), and the 0% of the scalar heads reflects models that score low on everything, not models that reject the edit.

8 Generalization

The original limitations section notes that no out-of-domain split was tested. We add one. FRJUDGE clauses come from identifiable source documents, which we recover from the raw export. Random splitting therefore lets clauses from the same insurance form appear in both training and test, and insurance forms are highly templated. We compare the standard random split against a document-grouped split in which no source document straddles the boundary (Table 9); the difference estimates how much of the reported performance is form-specific. We rerun three of the seven configurations of Table 8 on that split: the scalar baseline and the two augmented mixtures. The remaining four differ from the baseline in what they are trained to predict rather than in what they can exploit about a source form, so the leakage they are exposed to is the same, and running them again would cost training time without addressing the question the split is asked to answer.

The corpus makes that harder than it sounds. Its 297 pairs come from only 21 source forms and the three largest hold 71% of them, so the obvious construction, sweeping the forms in random order and cutting at the cumulative 60/70% marks, lets a single large form overshoot a boundary and starve the split behind it. On seeds 42–46 that procedure returns test splits of 16, 87, 30, 0 and 88 pairs, one of them empty and one far too small to carry a correlation. We therefore choose the form-to-split assignment by a short local search over total deviation from the target sizes. Thousands of assignments hit $178/30/89$ exactly, so each seed still selects a genuinely different partition while every split matches the random-split sizes of Table 8 pair for pair, and the two tables differ only in which forms are held out. One limitation is structural rather than incidental: the largest form is 37% of the corpus, larger than the 30% test quota, so it always falls in training and the grouped test set is always drawn from the remaining 188 pairs.

The comparison does not resolve. The scalar baseline's correlation falls from 0.542 on the random split to 0.454 on the grouped one, a drop of 0.088 that is in the direction leakage would predict. It does not

Model	r	ρ	RMSE	over%	ident. pass	unrel. pass	LEGALEDIT false-acc.	LEGALEDIT margin
JudgeBERT-Scalar (baseline)	$0.454_{\pm 0.14}$	0.282	2.49	24.0	0	0	0	–
JudgeBERT-DA	$0.190_{\pm 0.21}$	0.140	2.63	33.0	0	100	9	0.004
JudgeBERT-DA+LegalEdit	$0.149_{\pm 0.13}$	0.319	2.65	28.1	0	49	0	0.030
NLI-bidirectional (untrained)	0.391	0.401	2.20	45.7	1	–	–	*0.670*
Human ceiling	**0.597**	0.620	3.27	–	–	–	–	–

Table 9: Document-grouped split: no source form appears in both training and test. Columns are as in Table 8, and the splits are size-matched to it (178/30/89). The untrained NLI reference is recomputed on the grouped splits rather than carried over, since a corpus-level correlation is not comparable to a document-grouped one; the human ceiling is a property of the annotations and does not depend on the split. Comparing against Table 8 isolates the part of the reported correlation that survives a change of source document.

Metric	r within tercile of $\lvert\,\lvert s\rvert - \lvert o\rvert\,\rvert$: small	medium	large	all
$-\lvert\,\lvert s\rvert - \lvert o\rvert\,\rvert$ (length difference)	0.243	0.165	0.585	0.613
BERTScore-FlauBERT	0.198	0.449	0.413	0.484
LaBSE	0.174	0.532	0.240	0.424
NLI-bidirectional (min)	0.058	0.269	0.291	0.365
Surface-Ridge (10 features)	0.118	0.305	0.540	0.620
JudgeBERT-Scalar (baseline)	0.042	0.149	0.602	0.496

Table 10: Correlation with the human label *within* terciles of the absolute length change, pooling test predictions over seeds. Restricting to a tercile removes most of the length variance; a metric that measures legal meaning should degrade far less than one that measures deletion.

survive inspection. The per-seed differences are $+0.169$, -0.097, -0.416, -0.182 and $+0.086$: two of the five seeds *improve* under the harder split, and the grouped spread (± 0.143) is wider than the gap itself. The two augmented mixtures move by -0.002 and $+0.026$, which is to say not at all. As a control, the untrained NLI reference is fitted to nothing and should be indifferent to how the corpus is partitioned; it moves from 0.395 to 0.391, confirming that the two splits are otherwise comparable and that the variance above is the models' and not the split's.

We therefore do not claim that document-level leakage inflates the reported correlation, and we would ask that our own evidence be held to the standard §5 applies to the original. What we can say is narrower and, we think, more useful to anyone planning a successor corpus: FRJUDGE is too small and too concentrated to answer the question. Its grouped test split is always drawn from the 188 pairs the largest form leaves behind, at which size the seed-to-seed variation swamps the effect being measured. A corpus intended to support an out-of-domain claim needs its documents spread far more evenly than three forms holding 71% of the data, and this is the aspect of the design we would change first.

We further break the test predictions down by the magnitude of the length change between original and simplification. If a metric's correlation is concentrated in the buckets where the simplification is much shorter, it is detecting deletion rather than legal change, the hypothesis raised in §5. Table 10 reports Pearson r inside each tercile of $\lvert\,\lvert s\rvert - \lvert o\rvert\,\rvert$. Within a tercile the length signal is largely held constant, so a metric that retains its correlation there is using something other than deletion; one whose correlation collapses was reading length all along.

The result qualifies §5 in a way that is worth stating carefully, because it cuts against our own argument.

The length feature's correlation is overwhelmingly a *between*-tercile effect: pooled over the corpus it reaches 0.613, but inside the small and medium terciles it falls to 0.243 and 0.165. Once items of comparable length change are compared with each other, knowing the length change tells you little. It recovers to 0.585 only in the tercile where the simplification is much shorter than the original, which is exactly where deletion is available to be detected. The neural baselines have the opposite profile: in the medium tercile BERTSCORE over FlauBERT reaches 0.449 and LaBSE 0.532, at or above their pooled values of 0.484 and 0.424, where the length feature retains almost nothing. The supervised surface ridge tracks the raw length feature bucket for bucket, which is what one would expect if it is largely a length model with extra terms. The small tercile is hard for everything, which we read as a floor effect rather than as a result: when the two sentences are close to the same length, none of these signals has much to go on.

The honest reading is therefore narrower than "legal meaning preservation is length preservation". The corpus-level ranking is dominated by length, and any comparison that omits a length control, as the original does, will attribute to a metric what it should attribute to that confound. But the neural metrics are not *only* reading length. Where the length signal is weakest, they are the ones that hold up, keeping or exceeding the correlation they had over the whole corpus; whatever they are reading in that stratum, it is not the length change. What §6 adds is that the residual signal they retain is still not legal sensitivity, since the same metrics move almost not at all when the law is changed and the length is held exactly fixed. The two analyses answer different questions, and a metric has to survive both.

9 Discussion

What the sanity checks should be. We propose that a metric claiming to measure legal meaning report three checks, not two: identical pairs (does it saturate correctly?), unrelated pairs (does it bottom out?), and LEGALEDIT (does it move when, and only when, the law moves?). The first two are jointly satisfiable by lexical overlap and so cannot distinguish a legal metric from a similarity function; the third cannot be passed by overlap at all. That LEGALEDIT is passable matters as much as that it is hard: because bidirectional NLI clears it while every similarity metric does not, a low margin identifies a specific, fixable deficiency rather than an inherent limit of automatic evaluation.

We add a second recommendation, which our results force on us. The three checks should be reported *jointly and separately*, and none of them should be used as an admission criterion. In this study the metric that scores highest on the diagnostic is also the metric that fails the identical-pair check most badly, and a pipeline that screened candidates on saturation, as the current convention invites, would have discarded it before the interesting question was asked. Saturation, bottoming out, and legal sensitivity are three independent properties; a scorer can be repaired for the first two by calibration, and cannot be repaired for the third at all. Rank on the property that cannot be fixed. We also caution that a metric trained on its own sanity checks has not thereby passed them, and our runs show that caution cutting harder than we expected. The original reports JUDGEBERT-DA at 100% on both checks, with those exact distributions in its training mixture. Our augmented models pass the unrelated check on 77% and 69% of probes and the identical check on none of them (Table 8), and the difference between those two outcomes is only whether the probe comes from the same documents as the training pairs. Training on a constraint buys satisfaction of it in distribution and nothing at all across a change of source. The honest report separates "satisfies the constraint by construction" from "generalises to the constraint", and only the second is evidence about a metric.

What to report alongside correlation. Given a corpus with k ratings per item, the human ceiling is free to compute and should be mandatory. We recommend reporting (i) leave-one-annotator-out correlation as the "as good as an expert" bound, (ii) the Spearman–Brown reliability of the aggregate as the "as good as the aggregate allows" bound, (iii) a trivial surface-feature baseline, and (iv) all metrics calibrated to the human scale under identical supervision. Each of these changed a conclusion in this study.

Entailment is the better starting point. The construct FRJUDGE annotates is defined by an error taxonomy whose two dominant categories are omission and hallucination. Those are precisely the two directions of entailment failure, and our results suggest the correspondence is not merely conceptual: the only scorer that reacts to a legally decisive edit is the one built from bidirectional entailment. We would not conclude that an off-the-shelf multilingual NLI model is an adequate legal metric: its correlation with the human label is middling ($r = 0.395$), it is slower than a fine-tuned encoder, and its margin of 0.67 still leaves a third of the range unspent. The conclusion we draw is narrower and, we think, more useful: the next legal-meaning metric should be built on an entailment objective and evaluated on the diagnostic, rather than on a similarity objective selected by corpus correlation. A supervised model trained against FRJUDGE *and* minimally-edited negatives is the obvious candidate, and §7 reports a first attempt.

Disagreement is the finding, not the noise. That two of five Quebec law students systematically rate legal preservation three points lower than the other three is not an annotation defect to be averaged away. It is a measurement of how much legal interpretation of an isolated clause varies among trained readers, and any deployed metric that returns a single number conceals it. A metric that reports "6.2, experts ranged 2–9" tells a practitioner something actionable, namely that this clause needs a human, which "6.2" does not.

10 Reproducibility

Every number in this paper is produced by a single pipeline that runs end to end on a laptop; the largest model involved is CamemBERTv2-base (112M parameters) and no GPU cluster is required. We release the experiment code, the reconstruction of FRJUDGE from the raw export (which recovers annotator identity and source document, both dropped from the public JSONL), the document-grouped splits, the LEGALEDIT generator, and the result files behind every table.

One caveat on exact reproduction: our training runs use Apple MPS kernels, which are not bitwise deterministic, so a fixed seed reproduces a run only approximately: repeat runs of the scalar configuration at seed 42 differ in test-set r by several hundredths, which is comparable to the spread across seeds. We therefore report means and standard deviations over five seeds and treat any difference smaller than that spread as unresolved, the same discipline §5 argues the original comparison needs, applied to ourselves. Anyone reproducing this on CUDA should expect tighter agreement and correspondingly narrower error bars.

Two further details matter for anyone reproducing the diagnostic. The first is what the probes share with training, which differs by regime and which we state precisely because the original study's conflation of the two is one of our criticisms. The *identical* probe is built from statutory sentences (the Quebec Automobile Insurance Act and the Highway Safety Code) while the identical pairs in the -DA training mixture are built from FRJUDGE's insurance-form clauses, so a model passes that probe only by generalising across source documents. The LEGALEDIT probe is likewise generated from statutory sentences, while the minimally-edited negatives added by the DA+LEGALEDIT variant are generated from FRJUDGE originals: same perturbation rules, disjoint sentences. The *unrelated* probe is the weakest of the three in this respect: both it and the unrelated pairs in the training mixture are drawn from the same two statutes, under different random seeds, so a high unrelated pass rate for a -DA model should be read as in-distribution and we do not treat it as evidence of generalisation. Second, the ordinal Krippendorff coefficient we report is computed with the same library and the same reliability matrix as the nominal one, so the difference between 0.104 and 0.325 is attributable to the distance function alone.

11 Limitations

Our LEGALEDIT labels are definitional rather than annotated: we assert that changing *doit* to *peut* changes legal meaning, rather than having lawyers rate each perturbation. We think this is defensible for tier-1 edits, which alter modality, party, polarity or quantum, and we report tier-2 scope shifts separately for exactly this reason; but a human-validated version would be stronger, and a small expert study is the obvious next step. The perturbations are also single-edit and template-generated, so they probe a specific and narrow kind of legal change; a system could in principle be tuned to them without acquiring general legal sensitivity, which is why we treat LEGALEDIT as a necessary condition and not a benchmark to be optimised.

That caveat bears directly on our most encouraging result, and we would rather state it than let a reader find it. Our perturbations are largely negation, modality, quantifier and antonym contrasts, which is close to the inventory natural language inference training data is built around. A model fine-tuned on XNLI has seen a great many *ne ... pas* insertions and *tous*/*certains* alternations, so bidirectional NLI's margin of 0.67 may reflect familiarity with the *form* of our edits rather than sensitivity to their legal consequence. The two hypotheses make different predictions: only the second should survive perturbations that change legal force without changing logical form, such as substituting a defined term for a near-synonym that is not its legal equivalent. Separating them needs a version of the diagnostic built around defined-term substitution, which our rule inventory reaches only through the single *automobile*/*véhicule* case. We report the NLI result as a promising direction with an identified confound, not as evidence that entailment models read law.

A further mismatch is that the diagnostic and the correlation evaluation run on different text. FRJUDGE is drawn from insurance forms, while the LEGALEDIT probes are generated from two Quebec statutes, because the statutes supply the volume and the regularity of phrasing that the perturbation rules need. The two are the same language and register but not the same genre, so Table 6 and Table 3 answer different questions about the same metric and should not be read as one ranking.

We do not evaluate an LLM-as-judge baseline, and this is the most significant gap in our comparison. Recent legal evaluation work [35] suggests that a prompted frontier model given the annotators' own rubric is the natural competitor to a fine-tuned 112M-parameter encoder, and a demonstration that the small supervised model matches it would be a strong efficiency result. We note two reasons the omission does not affect our conclusions. Our claims are about what the *published* comparison establishes and about a diagnostic property, sensitivity to a legally decisive edit, that is a necessary condition for any scorer, prompted or fine-tuned; and our evaluation harness takes an arbitrary scoring function, so LEGALEDIT can be run against a prompted judge without modification. We release the harness with that interface so the comparison can be completed.

Like the original work, we evaluate isolated clauses out of document context, and all simplifications come from a single generator (`gpt-4-turbo`) with one prompt. A metric meta-evaluation properly requires outputs from several systems spanning a range of quality; with one generator, a metric may be learning that generator's characteristic failure modes. Our generalization analysis addresses document-level leakage but cannot address generator-level overfitting.

Finally, our human ceiling is computed on all 297 items while the model correlations are computed on 89-item test splits, and the ceiling carries its own sampling uncertainty (per-annotator 95% CIs in Table 2). The comparison "model $>$ ceiling" should be read with those intervals in view.

12 Ethical Considerations

Annotator de-anonymisation in the released artefacts. The original paper pseudonymises its five annotators as A–E. The released raw export, however, records their real first names in the `_annotator_id` and `_session_id` fields, joined to per-annotation Unix timestamps and free-text justifications. This makes five named law students' individual legal judgments, working pace, and reasoning publicly attributable, contrary to the pseudonymisation the paper describes. We use only pseudonyms A–E throughout this work, our released code pseudonymises on load, and we do not reproduce the names. We have not been able to determine whether the annotators' consent covered named release; we flag it so that the maintainers can repair the file and, if appropriate, its history. We report this as a reproducibility and data-protection issue, not as a criticism of the study's conduct: releasing raw annotation exports is good practice, and this is a foreseeable and easily corrected side effect of it.

Use of the metric. Our results strengthen the original paper's own caution. A metric that cannot distinguish *doit* from *peut* must not be used as an automated gate on whether a simplified clause is safe to publish. We regard the appropriate deployment as triage, ranking clauses for human review and flagging disagreement, and not certification. Nothing here constitutes legal advice.

Corrigenda to the original paper. For the record, and offered as errata rather than criticism: Table 7 reports "Top K 0.9", but 0.9 is a `top_p` value and the OpenAI API exposes no `top_k`; §3.2.2 describes Kandel and Moles (1958) as a "Flesch-Kincaid grade level" with ≤ 50 indicating harder text and a discarded example at 69.87, which identifies it as French Flesch *Reading Ease* (higher = easier), not a grade level; §6.2 says "only three metrics" and lists four; Table 5's fourth column is headed `%>1%` while the text defines it as the ratio scoring at or below 1%; and SBERT-Multi's identical-pair score differs between the DA-false (0.00) and DA-true (100.00) blocks although it is an untrained metric whose behaviour on identical pairs cannot depend on how JUDGEBERT was trained. The repository README additionally documents a schema from a different project (a binary grammaticality label), and the dataset license differs between the paper (CC-BY 4.0) and the repository (CC-BY-NC-SA 4.0).

13 Conclusion

Legal meaning preservation is the right construct and FRJUDGE is a valuable corpus; both deserve to be built on. But the evidence currently offered for JUDGEBERT (a correlation above the human ceiling, against uncalibrated baselines, with no trivial control, on sanity checks it was trained on) does not establish that it measures legal meaning rather than lexical overlap and length. The decisive test is cheap: hold surface form fixed and move the law. On that test every similarity-based metric we examined, the state of the art included, scores a sentence that says the opposite of the original as though nothing had changed, and reversing which party the clause binds is among the least detectable edits of all. The test is not unpassable: bidirectional entailment, the baseline the original comparison omits, clears most of it. That is the encouraging half of the result, and it points at what the next version of this metric should be made of. It also carries a warning for how such metrics are selected, because the signal that ranks best on the corpus ranks last on the law.

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